Steering System

The robot will use an **Ackermann steering mechanism** to improve turning efficiency by allowing the inner and outer front wheels to follow different turning radii. This reduces tyre scrub during cornering, resulting in smoother turns, improved stability, and more consistent path tracking.

The steering linkage will be designed to minimise mechanical play while remaining easy to assemble, adjust, and replace when required. The design will also prioritise compactness and low weight to remain consistent with the robot's overall design philosophy.

Steering will be actuated using the **EMAX ES08MA II Metal Gear Micro Servo**. The servo has been selected for its compact size, low weight, metal gear construction, adequate torque, and precise positioning. These characteristics make it suitable for the steering system of a lightweight autonomous vehicle, where both space and mechanical accuracy are important.

Components

- EMAX ES08MA II Metal Gear Micro Servo
- Front Wheel Steering Assembly
- Ackermann Steering Linkage

Propulsion System

The robot will use a **rear-wheel-drive (RWD)** configuration powered by a single **N20 metal gear motor with an encoder**.

The RWD layout simplifies the drivetrain while keeping the steering and propulsion systems mechanically independent. It is also well suited to the robot's compact architecture and allows the front assembly to remain dedicated to steering.

The drivetrain will be evaluated using three possible transmission mechanisms.

1. Spur Gear Drive

Power will be transmitted from the motor to the rear axle using a pair of spur gears. This arrangement allows the gear ratio to be selected according to the required balance between speed and torque.

Advantages:

- Compact and lightweight.

- Simple to manufacture and assemble.
- Allows easy modification of the gear ratio.
- Suitable for direct integration into a compact chassis.

2. Timing Belt Drive

A timing belt and pulley system will transfer power from the motor to the rear axle. This provides greater flexibility in motor placement while maintaining positive engagement between the motor and axle.

Advantages:

- Flexible motor placement.
- Smooth power transmission.
- Positive engagement with minimal slipping.
- Can reduce the need for precise alignment between the motor and axle.

3. Rear Differential Drive

Power from the motor will be transmitted through a differential, allowing the left and right rear wheels to rotate at different speeds while cornering.

This configuration can improve cornering behaviour and reduce tyre scrub compared with a locked rear axle. However, it introduces additional mechanical components, weight, and potential points of failure.

Advantages:

- Allows different wheel speeds during cornering.
- Improves rear-wheel behaviour during turns.
- Reduces tyre scrub.

Disadvantages:

- Greater mechanical complexity.
- Higher component count.
- Increased weight.
- More difficult to manufacture and maintain at the robot’s scale.

The final propulsion mechanism will be selected based on the balance between **weight, efficiency, mechanical simplicity, reliability, and available space** within the chassis.